

4.3 Mixed and Entire Radicals



LESSON FOCUS

Express an entire radical as a mixed radical, and vice versa.

This quilt represents a Pythagorean spiral. The smallest triangle is a right isosceles triangle with legs 1 unit long.

Make Connections

We can name the fraction $\frac{3}{12}$ in many different ways:

$$\frac{1}{4} \quad \frac{5}{20} \quad \frac{30}{120} \quad \frac{100}{400}$$

How do you show that each fraction is equivalent to $\frac{3}{12}$?

Why is $\frac{1}{4}$ the simplest form of $\frac{3}{12}$?

Construct Understanding

TRY THIS

Work with a partner.

You will need 1-cm grid paper and a calculator.

- On grid paper, draw an isosceles right triangle with legs 1 cm long. Write the length of the hypotenuse as a radical. Label the lengths of the sides on the triangle.
- Draw an isosceles right triangle with legs 2 cm long. Write the length of the hypotenuse as a radical. Label the lengths of the sides on the triangle.

- C.** Explain why the triangle in Step B is an enlargement of the triangle in Step A.
What is the scale factor of the enlargement?
How is the length of the hypotenuse in the larger triangle related to the corresponding length in the smaller triangle?
- D.** Draw isosceles right triangles with legs: 3 cm long; 4 cm long; and 5 cm long. For each triangle, write the length of the hypotenuse in 2 different ways.
- E.** Describe any relationships in the lengths of the sides of the triangles. Which form of the radical makes the relationships easier to see?

Just as with fractions, equivalent expressions for any number have the same value.

- $\sqrt{16 \cdot 9}$ is equivalent to $\sqrt{16} \cdot \sqrt{9}$ because:

$$\begin{array}{rcl} \sqrt{16 \cdot 9} & = & \sqrt{144} \\ & = & 12 \end{array} \quad \text{and} \quad \begin{array}{rcl} \sqrt{16} \cdot \sqrt{9} & = & 4 \cdot 3 \\ & = & 12 \end{array}$$

- Similarly, $\sqrt[3]{8 \cdot 27}$ is equivalent to $\sqrt[3]{8} \cdot \sqrt[3]{27}$ because:

$$\begin{array}{rcl} \sqrt[3]{8 \cdot 27} & = & \sqrt[3]{216} \\ & = & 6 \end{array} \quad \text{and} \quad \begin{array}{rcl} \sqrt[3]{8} \cdot \sqrt[3]{27} & = & 2 \cdot 3 \\ & = & 6 \end{array}$$

Multiplication Property of Radicals

$$\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b},$$

where n is a natural number, and a and b are real numbers

We can use this property to simplify square roots and cube roots that are not perfect squares or perfect cubes, but have factors that are perfect squares or perfect cubes.

For example, the factors of 24 are: 1, 2, 3, 4, 6, 8, 12, and 24.

- We can simplify $\sqrt{24}$ because 24 has a perfect square factor of 4. Rewrite 24 as the product of two factors, one of which is 4.

$$\begin{aligned} \sqrt{24} &= \sqrt{4 \cdot 6} \\ &= \sqrt{4} \cdot \sqrt{6} \\ &= 2 \cdot \sqrt{6} \\ &= 2\sqrt{6} \end{aligned}$$

We read $2\sqrt{6}$ as “2 root 6.”

- Similarly, we can simplify $\sqrt[3]{24}$ because 24 has a perfect cube factor of 8. Rewrite 24 as the product of two factors, one of which is 8.

$$\begin{aligned}\sqrt[3]{24} &= \sqrt[3]{8 \cdot 3} \\ &= \sqrt[3]{8} \cdot \sqrt[3]{3} \\ &= 2 \cdot \sqrt[3]{3} \\ &= 2\sqrt[3]{3}\end{aligned}$$

We read $2\sqrt[3]{3}$ as “2 cube root 3.”

- However, we cannot simplify $\sqrt[4]{24}$ because 24 has no factors (other than 1) that can be written as a fourth power.

We can also use prime factorization to simplify a radical.

Example 1

Simplifying Radicals Using Prime Factorization

Simplify each radical.

a) $\sqrt{80}$ b) $\sqrt[3]{144}$ c) $\sqrt[4]{162}$

SOLUTION

Write each radical as a product of prime factors, then simplify.

a) $\sqrt{80} = \sqrt{8 \cdot 10}$

$$\begin{aligned}&= \sqrt{2 \cdot 2 \cdot 2 \cdot 5 \cdot 2} \\ &= \sqrt{(2 \cdot 2) \cdot (2 \cdot 2) \cdot 5} \\ &= \sqrt{2 \cdot 2} \cdot \sqrt{2 \cdot 2} \cdot \sqrt{5} \\ &= 2 \cdot 2 \cdot \sqrt{5} \\ &= 4\sqrt{5}\end{aligned}$$

Since $\sqrt{80}$ is a square root, look for factors that appear twice.

b) $\sqrt[3]{144} = \sqrt[3]{12 \cdot 12}$

$$\begin{aligned}&= \sqrt[3]{2 \cdot 2 \cdot 3 \cdot 2 \cdot 2 \cdot 3} \\ &= \sqrt[3]{(2 \cdot 2 \cdot 2) \cdot 2 \cdot 3 \cdot 3} \\ &= \sqrt[3]{2 \cdot 2 \cdot 2} \cdot \sqrt[3]{2 \cdot 3 \cdot 3} \\ &= 2 \cdot \sqrt[3]{2 \cdot 3 \cdot 3} \\ &= 2\sqrt[3]{18}\end{aligned}$$

Since $\sqrt[3]{144}$ is a cube root, look for factors that appear 3 times.

c) $\sqrt[4]{162} = \sqrt[4]{81 \cdot 2}$

$$\begin{aligned}&= \sqrt[4]{3 \cdot 3 \cdot 3 \cdot 3 \cdot 2} \\ &= \sqrt[4]{(3 \cdot 3 \cdot 3 \cdot 3) \cdot 2} \\ &= \sqrt[4]{3 \cdot 3 \cdot 3 \cdot 3} \cdot \sqrt[4]{2} \\ &= 3\sqrt[4]{2}\end{aligned}$$

Since $\sqrt[4]{162}$ is a fourth root, look for factors that appear 4 times.

CHECK YOUR UNDERSTANDING

1. Simplify each radical.

a) $\sqrt{63}$

b) $\sqrt[3]{108}$

c) $\sqrt[4]{128}$

[Answers: a) $3\sqrt{7}$ b) $3\sqrt[3]{4}$ c) $2\sqrt[4]{8}$]

Some numbers, such as 200, have more than one perfect square factor.
 The factors of 200 are: 1, 2, 4, 5, 8, 10, 20, 25, 40, 50, 100, 200
 Since 4, 25, and 100 are perfect squares, we can simplify $\sqrt{200}$ in these ways.

$$\begin{array}{lll} \sqrt{200} = \sqrt{4 \cdot 50} & \sqrt{200} = \sqrt{25 \cdot 8} & \sqrt{200} = \sqrt{100 \cdot 2} \\ = \sqrt{4} \cdot \sqrt{50} & = \sqrt{25} \cdot \sqrt{8} & = \sqrt{100} \cdot \sqrt{2} \\ = 2\sqrt{50} & = 5\sqrt{8} & = 10\sqrt{2} \end{array}$$

$10\sqrt{2}$ is in simplest form because the radical contains no perfect square factors other than 1.

To write a radical of index n in simplest form, we write the radicand as a product of 2 factors, one of which is the greatest perfect n th power.

Example 2 Writing Radicals in Simplest Form

Write each radical in simplest form, if possible.

a) $\sqrt[3]{40}$ b) $\sqrt{26}$ c) $\sqrt[4]{32}$

SOLUTION

Look for perfect n th factors, where n is the index of the radical.

- a) The factors of 40 are: 1, 2, 4, 5, 8, 10, 20, 40
 The greatest perfect cube is $8 = 2 \cdot 2 \cdot 2$,
 so write 40 as $8 \cdot 5$.

$$\begin{aligned} \sqrt[3]{40} &= \sqrt[3]{8 \cdot 5} \\ &= \sqrt[3]{8} \cdot \sqrt[3]{5} \\ &= 2 \cdot \sqrt[3]{5} \\ &= 2\sqrt[3]{5} \end{aligned}$$

- b) The factors of 26 are: 1, 2, 13, 26
 There are no perfect square factors other than 1.
 So, $\sqrt{26}$ cannot be simplified.

- c) The factors of 32 are: 1, 2, 4, 8, 16, 32
 The greatest perfect fourth power is $16 = 2 \cdot 2 \cdot 2 \cdot 2$,
 so write 32 as $16 \cdot 2$.

$$\begin{aligned} \sqrt[4]{32} &= \sqrt[4]{16 \cdot 2} \\ &= \sqrt[4]{16} \cdot \sqrt[4]{2} \\ &= 2 \cdot \sqrt[4]{2} \\ &= 2\sqrt[4]{2} \end{aligned}$$

CHECK YOUR UNDERSTANDING

2. Write each radical in simplest form, if possible.

a) $\sqrt{30}$

b) $\sqrt[3]{32}$

c) $\sqrt[4]{48}$

[Answers: a) cannot be simplified
 b) $2\sqrt[3]{4}$ c) $2\sqrt[4]{3}$]

Radicals of the form $\sqrt[n]{x}$ such as $\sqrt{80}$, $\sqrt[3]{144}$, and $\sqrt[4]{162}$ are **entire radicals**.

Radicals of the form $a\sqrt[n]{x}$ such as $4\sqrt{5}$, $2\sqrt[3]{18}$, and $3\sqrt[4]{2}$ are **mixed radicals**. Entire radicals were rewritten as mixed radicals in *Examples 1* and *2*.

Any number can be written as the square root of its square; for example, $2 = \sqrt{2 \cdot 2}$, $3 = \sqrt{3 \cdot 3}$, $4 = \sqrt{4 \cdot 4}$, and so on. Similarly, any number can be written as the cube root of its cube, or the fourth root of its perfect fourth power. We use this strategy to write a mixed radical as an entire radical.

Example 3 Writing Mixed Radicals as Entire Radicals

Write each mixed radical as an entire radical.

a) $4\sqrt{3}$ b) $3\sqrt[3]{2}$ c) $2\sqrt[5]{2}$

SOLUTION

a) Write 4 as: $\sqrt{4 \cdot 4} = \sqrt{16}$

$$\begin{aligned} 4\sqrt{3} &= \sqrt{16} \cdot \sqrt{3} && \text{Use the Multiplication Property of Radicals.} \\ &= \sqrt{16 \cdot 3} \\ &= \sqrt{48} \end{aligned}$$

b) Write 3 as: $\sqrt[3]{3 \cdot 3 \cdot 3} = \sqrt[3]{27}$

$$\begin{aligned} 3\sqrt[3]{2} &= \sqrt[3]{27} \cdot \sqrt[3]{2} \\ &= \sqrt[3]{27 \cdot 2} \\ &= \sqrt[3]{54} \end{aligned}$$

c) Write 2 as: $\sqrt[5]{2 \cdot 2 \cdot 2 \cdot 2 \cdot 2} = \sqrt[5]{32}$

$$\begin{aligned} 2\sqrt[5]{2} &= \sqrt[5]{32} \cdot \sqrt[5]{2} \\ &= \sqrt[5]{32 \cdot 2} \\ &= \sqrt[5]{64} \end{aligned}$$

CHECK YOUR UNDERSTANDING

3. Write each mixed radical as an entire radical.

a) $7\sqrt{3}$

b) $2\sqrt[3]{4}$

c) $2\sqrt[5]{3}$

[Answers: a) $\sqrt{147}$ b) $\sqrt[3]{32}$ c) $\sqrt[5]{96}$]

How would rewriting mixed radicals as entire radicals help you to order a set of mixed radicals with the same index?

Discuss the Ideas

1. How can you determine if an entire radical can be written as a mixed radical?
2. Suppose an entire radical can be simplified. How do you use the Multiplication Property of Radicals to write it in simplest form?

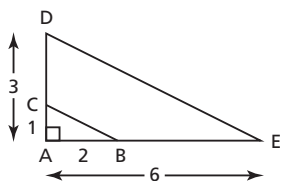
Exercises

A

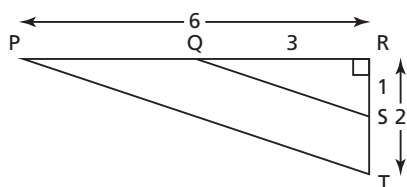
- List all the perfect squares up to 400, and their square roots.
- Write each radical in simplest form.
 - $\sqrt{8}$
 - $\sqrt{12}$
 - $\sqrt{32}$
 - $\sqrt{50}$
 - $\sqrt{18}$
 - $\sqrt{27}$
 - $\sqrt{48}$
 - $\sqrt{75}$
- Write each mixed radical as an entire radical.
 - $5\sqrt{2}$
 - $6\sqrt{2}$
 - $7\sqrt{2}$
 - $8\sqrt{2}$
 - $5\sqrt{3}$
 - $6\sqrt{3}$
 - $7\sqrt{3}$
 - $8\sqrt{3}$
- List all the perfect cubes up to 1000, and their cube roots.
 - List all the perfect fourth powers up to 1000, and their fourth roots.

B

- Use the diagram to explain why $\sqrt{45} = 3\sqrt{5}$.



- Use algebra to verify that $\sqrt{45} = 3\sqrt{5}$.
- Use the diagram to explain why $\sqrt{40} = 2\sqrt{10}$.



- Use algebra to verify that $\sqrt{40} = 2\sqrt{10}$.
- Explain why rewriting $\sqrt{50}$ as $\sqrt{25} \cdot \sqrt{2}$ helps you simplify $\sqrt{50}$, but rewriting $\sqrt{50}$ as $\sqrt{10} \cdot \sqrt{5}$ does not.
- Write each radical in simplest form, if possible.
 - $\sqrt{90}$
 - $\sqrt{73}$
 - $\sqrt{108}$
 - $\sqrt{600}$
 - $\sqrt{54}$
 - $\sqrt{91}$
 - $\sqrt{28}$
 - $\sqrt{33}$
 - $\sqrt{112}$

- Write each radical in simplest form, if possible.

- | | |
|--------------------|--------------------|
| a) $\sqrt[3]{16}$ | b) $\sqrt[3]{81}$ |
| c) $\sqrt[3]{256}$ | d) $\sqrt[3]{128}$ |
| e) $\sqrt[3]{60}$ | f) $\sqrt[3]{192}$ |
| g) $\sqrt[3]{135}$ | h) $\sqrt[3]{100}$ |
| i) $\sqrt[3]{500}$ | j) $\sqrt[3]{375}$ |

- Write each mixed radical as an entire radical.

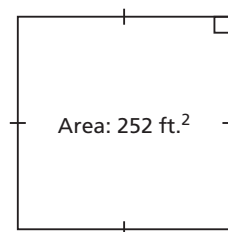
- | | |
|-------------------|-------------------|
| a) $3\sqrt{2}$ | b) $4\sqrt{2}$ |
| c) $6\sqrt{5}$ | d) $5\sqrt{6}$ |
| e) $7\sqrt{7}$ | f) $2\sqrt[3]{2}$ |
| g) $3\sqrt[3]{3}$ | h) $4\sqrt[3]{3}$ |
| i) $5\sqrt[3]{2}$ | j) $2\sqrt[3]{9}$ |

- Can every mixed radical be expressed as an entire radical?

- Can every entire radical be expressed as a mixed radical?

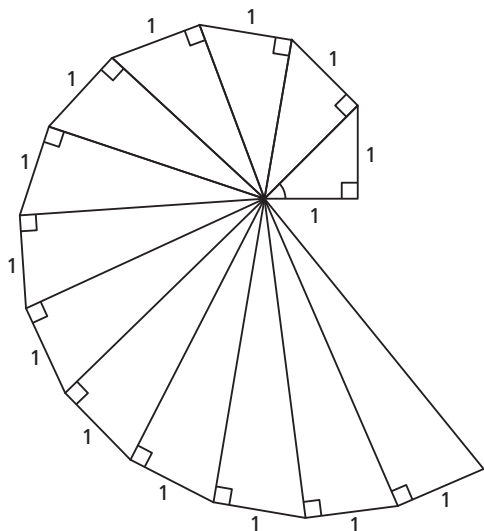
Give examples to support your answers.

- Express the side length of this square as a radical in simplest form.



- A cube has a volume of 200 cm^3 . Write the edge length of the cube as a radical in simplest form.
- A square has an area of 54 square inches. Determine the perimeter of the square. Write the answer as a radical in simplest form.
- Write each radical in simplest form.
 - $\sqrt[4]{48}$
 - $\sqrt[4]{405}$
 - $\sqrt[4]{1250}$
 - $\sqrt[4]{176}$
- Write each mixed radical as an entire radical.
 - $6\sqrt[4]{3}$
 - $7\sqrt[4]{2}$
 - $3\sqrt[5]{4}$
 - $4\sqrt[5]{3}$

19. The quilt on page 213 is made from right triangles. In Chapter 2, page 77, you determined the tangents of the angles at the centre of the spiral. The first triangle is a right isosceles triangle with legs 1 unit long. The hypotenuse of this triangle is one leg of the second triangle, with its other leg 1 unit long. This pattern continues.



- a) Calculate the length of the hypotenuse of each triangle. Write each length as an entire radical.
- b) i) What pattern do you see in the lengths?
 ii) Use this pattern to predict the length of the hypotenuse of the 50th triangle.
 iii) How many of the first 100 triangles have hypotenuse lengths that can be written as mixed radicals? Justify your answer.

20. Here is a student's solution for writing $8\sqrt[3]{2}$ as an entire radical.

$$\begin{aligned} 8\sqrt[3]{2} &= 8 \cdot \sqrt[3]{2} \\ &= \sqrt[3]{2} \cdot \sqrt[3]{2} \\ &= \sqrt[3]{2 \cdot 2} \\ &= \sqrt[3]{4} \end{aligned}$$

Identify an error the student made, then write the correct solution.

21. A student simplified $\sqrt{96}$ as shown:

$$\begin{aligned} \sqrt{96} &= \sqrt{4} \cdot \sqrt{48} \\ &= 2 \cdot \sqrt{48} \\ &= 2 \cdot \sqrt{8} \cdot \sqrt{6} \\ &= 2 \cdot 4 \cdot \sqrt{6} \\ &= 8\sqrt{6} \end{aligned}$$

Identify the errors the student made, then write a correct solution.

22. Arrange in order from greatest to least. What strategy did you use each time?

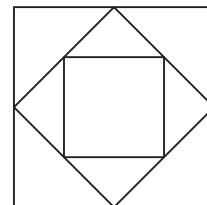
- a) $9\sqrt{2}$, $2\sqrt{6}$, $8\sqrt{3}$, $4\sqrt{5}$, $6\sqrt{2}$
 b) $4\sqrt{7}$, $8\sqrt{3}$, $2\sqrt{13}$, $6\sqrt{5}$
 c) $7\sqrt{3}$, $9\sqrt{2}$, $5\sqrt{6}$, $\sqrt{103}$, $3\sqrt{17}$

23. Simplify the radicals in each list. What patterns do you see in the results? Write the next 2 radicals in each list.

- | | |
|------------------|--------------------------|
| a) $\sqrt{4}$ | b) $\sqrt[3]{27}$ |
| $\sqrt{400}$ | $\sqrt[3]{27\,000}$ |
| $\sqrt{40\,000}$ | $\sqrt[3]{27\,000\,000}$ |
| c) $\sqrt{8}$ | d) $\sqrt[3]{24}$ |
| $\sqrt{800}$ | $\sqrt[3]{24\,000}$ |
| $\sqrt{80\,000}$ | $\sqrt[3]{24\,000\,000}$ |

C

24. The largest square in this diagram has side length 8 cm. Calculate the side length and area of each of the two smaller squares. Write the radicals in simplest form.



25. Given that $\sqrt{2} \doteq 1.4142$, determine a decimal approximation for each radical, without using a calculator.

- a) i) $\sqrt{200}$ ii) $\sqrt{20\,000}$
 b) i) $\sqrt{8}$ ii) $\sqrt{18}$ iii) $\sqrt{32}$ iv) $\sqrt{50}$

Reflect

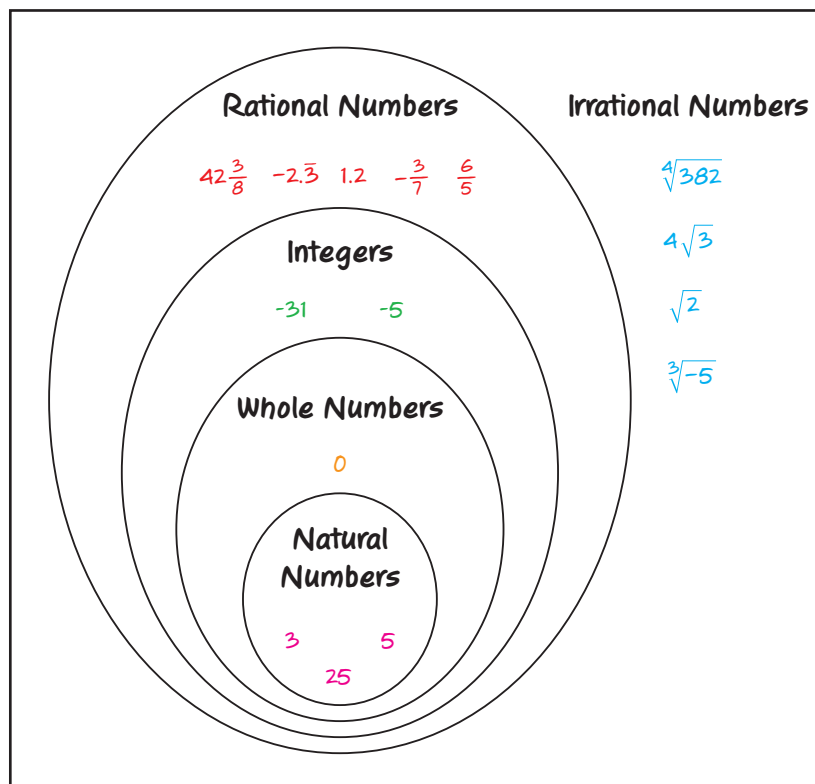
How do you use the index of a radical when you simplify a radical, and when you write a mixed radical as an entire radical? Use examples to support your explanation.



CHECKPOINT 1

Connections

Real Numbers



Concept Development

In Lesson 4.1

- You applied what you know about **square roots** to explore **decimal approximations of cube roots and fourth roots**.
- You determined that **some radicals** can be represented as **rational numbers** and other radicals cannot.

In Lesson 4.2

- You defined **irrational numbers**, and represented these numbers and **rational numbers** as the **set of real numbers**.
- You **identified** conditions for which a **radical** has a **rational number value**, and **estimated** the values of **radicals that are irrational**.

In Lesson 4.3

- You defined **mixed radicals** and **entire radicals**, and used factoring to simplify radicals.

Assess Your Understanding

4.1

1. Evaluate each radical. How did you use the index of the radical in your work?
a) $\sqrt{81}$ b) $\sqrt[3]{-125}$ c) $\sqrt[4]{256}$ d) $\sqrt[5]{243}$
2. Estimate the value of each radical to 2 decimal places.
How can you do this without using the root keys on a calculator?
a) $\sqrt{10}$ b) $\sqrt[3]{15}$ c) $\sqrt[4]{9}$ d) $\sqrt[5]{23}$
3. Does the decimal representation of $\sqrt[4]{60}$ repeat, terminate, or neither?
Justify your answer.

4.2

4. Tell whether each number is rational or irrational. Justify your answers.
a) $\sqrt{11}$ b) $\sqrt[3]{16}$ c) $\sqrt[3]{-16}$ d) $\sqrt{121}$ e) $\sqrt{\frac{121}{16}}$ f) $\sqrt{12.1}$
5. For each irrational number, sketch a number line and label its approximate location. Describe your strategies.
a) $\sqrt{19}$ b) $\sqrt[3]{-20}$ c) $\sqrt[4]{30}$ d) $\sqrt[3]{36}$
6. a) Draw a diagram to illustrate the real number system.
Write the numbers below in the appropriate places on your diagram.
i) $3\frac{1}{3}$ ii) -42 iii) 4.5 iv) $-4.\bar{5}$
v) 0 vi) 14 vii) $\sqrt{7}$ viii) π
b) Choose 1 more number for each section of your diagram, where possible. Write each number in the correct place on your diagram.
7. a) Sketch a number line and mark each number on it.
i) $\sqrt{32}$ ii) $\sqrt[3]{72}$ iii) $\sqrt[4]{100}$ iv) $\sqrt[3]{50}$ v) $\sqrt{65}$ vi) $\sqrt[4]{60}$
b) Order the numbers in part a from greatest to least.
8. Sketch a square. Label its area so that:
a) The perimeter of the square is a rational number.
b) The perimeter of the square is an irrational number.

4.3

9. Write each radical in simplest form, if possible.
a) $\sqrt{45}$ b) $\sqrt[3]{96}$ c) $\sqrt{17}$ d) $\sqrt[4]{48}$ e) $\sqrt[3]{80}$ f) $\sqrt[4]{50}$
10. Choose one radical from question 9 that can be simplified.
Write a set of instructions for simplifying the radical.
11. Rewrite each mixed radical as an entire radical.
a) $3\sqrt{7}$ b) $2\sqrt[3]{4}$ c) $7\sqrt{3}$ d) $2\sqrt[4]{12}$ e) $3\sqrt[3]{10}$ f) $6\sqrt{11}$